A position and momentum of 1 keV electron are simultaneously measured. If position is located within 10 nm then what is the percentage uncertainty in its momentum?

Given : $E = 1\ \text{keV} = 1000 \times 1.6 \times 10^{-19}\ \text{J}$, $\Delta x = 10 \times 10^{-9}\text{m}$

$\frac{\Delta p}{p} \times 100 = ?$

$$p = \sqrt{2mE} = \sqrt{2 \times 9.1 \times 10^{-31} \times 1000 \times 1.6 \times 10^{-19}} = 1.706 \times 10^{-23}\ \frac{\text{kg.m}}{\text{s}}$$

According to HUP, $\Delta x . \Delta p \geq \frac{h}{4\pi}$

$$\therefore \Delta p \geq \frac{h}{4\pi} \times \frac{1}{\Delta x} = \frac{6.63 \times 10^{-34}}{4 \times 3.142 \times 10 \times 10^{-9}}$$

$$\therefore \Delta p = 5.275 \times 10^{-27}\ \frac{\text{kg.m}}{\text{s}}$$

$$\therefore \frac{\Delta p}{p} \times 100 = \frac{5.275 \times 10^{-27}}{1.706 \times 10^{-23}} \times 100 = 0.0309\ \%$$

An electron has a speed of 400 m/s with uncertainty of 0.01%. Find the accuracy in its position.

Given : $v = 400\frac{\text{m}}{\text{s}}$, $\frac{\Delta v}{v} = \frac{0.01}{100}$, $m = 9.1 \times 10^{-31}\ \text{kg}$,

$h = 6.63 \times 10^{-34}, \text{J} - \text{s}$ $\Delta x = ?$

$$p = m\,v = 9.1 \times 10^{-31} \times 400 = 3.64 \times 10^{-28}\ \frac{\text{kg.m}}{\text{s}}$$

$$\Delta p = m\,\Delta v = m\,v\,\frac{\Delta v}{v}$$

$$= p \times \frac{\Delta v}{v} = 3.64 \times 10^{-28} \times \frac{0.01}{100} = 3.64 \times 10^{-32}\ \frac{\text{kg.m}}{\text{s}}$$

According to HUP, $\Delta x . \Delta p \geq \frac{h}{4\pi}$

$$\therefore \Delta x \geq \frac{h}{4\pi} \times \frac{1}{\Delta p} = \frac{6.63 \times 10^{-34}}{4 \times 3.142 \times 3.64 \times 10^{-32}} = 1.449 \times 10^{-3}\text{m}$$

An electron has a speed of 900 m/s with an accuracy of 0.001%. Calculate the uncertainty in the position of the electron.

Given : $v = 900\frac{m}{s}$, $\frac{\Delta v}{v} = \frac{0.001}{100}$, $m = 9.1 \times 10^{-31}$ kg,

$h = 6.63 \times 10^{-34}\ J - s$, $\Delta x = ?$

$$p = m\,v = 9.1 \times 10^{-31} \times 900 = 8.19 \times 10^{-28}\,\frac{kg.m}{s}$$

$$\Delta p = m\,\Delta v = m\,v\,\frac{\Delta v}{v}$$

$$= p \times \frac{\Delta v}{v} = 8.19 \times 10^{-28} \times \frac{0.001}{100} = 8.19 \times 10^{-33}\,\frac{kg.m}{s}$$

According to HUP, $\Delta x.\Delta p \geq \frac{h}{4\pi}$

$$\therefore \Delta x \geq \frac{h}{4\pi} \times \frac{1}{\Delta p} = \frac{6.63 \times 10^{-34}}{4 \times 3.142 \times 8.19 \times 10^{-33}} = 6.44 \times 10^{-3}m$$

The speed of an electron is measured to within an uncertainty of 2 x 10⁴ m/s. What is the minimum space required by the electron to be confined to an atom?

Given : $\Delta v = 2 \times 10^{4}\frac{m}{s}$, $m = 9.1 \times 10^{-31}$ kg,

$h = 6.63 \times 10^{-34}\ J - s$, $\Delta x = ?$

According to HUP, $\Delta x\,\Delta p = \Delta x\,m\,\Delta v \geq \frac{h}{4\pi}$

$$\therefore \Delta x \geq \frac{h}{4\pi} \times \frac{1}{m\,\Delta v} = \frac{6.63\times10^{-34}}{4\times3.142\times9.1\times10^{-31}\times2\times10^{4}}$$

$$\therefore \Delta x = 2.898 \times 10^{-9}m$$

An electron confined in a box of length 10^{-8} m. Calculate minimum uncertainty in its velocity.

Given : $\Delta x = 10^{-8} m$, $m = 9.1 \times 10^{-31}$ kg,

$h = 6.63 \times 10^{-34}$, $\Delta v = ?$

According to HUP, $\Delta x . \Delta p = \Delta x . m \, \Delta v \geq \frac{h}{4\pi}$

$$\therefore \Delta v \geq \frac{h}{4\pi} \times \frac{1}{m \, \Delta x} = \frac{6.63 \times 10^{-34}}{4 \times 3.142 \times 9.1 \times 10^{-31} \times 10^{-8}}$$

$\therefore \Delta v = 5797 \text{ m/s}$